Quantitative M/C SHOW-YOUR-WORK page. **Name** Angel Sanchez **ID** _____
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

1. Given: $n = 1.00 mol$ $T = 100°C = 373K$
$V = 10.0L$ $P = ?\ 3.06 atm$

$P(10.0) = (1.00)(.08206)(373)$

$T_K = 100 + 273$ $\frac{P10.0}{10.0} = \frac{30.6}{10.0}$

$P = 3.06 atm$

7. $\frac{380 mmHg}{1} \times \frac{1 atm}{760 mmHg} = 0.50 atm$

13. $M = \frac{\text{Moles of Solute}}{\text{kg of Solvent}}$ $M = \frac{2.00 mol\ ZnCl_2}{.500 kg\ H_2O} =$ 4.00 M

14. $V_1 = \frac{C_2 V_2}{C_1}$ $V_1 = \frac{(6.100)(500.0) mL}{1.00} = \frac{50}{1.00} = 50 mL$

17. $1.00 atm = 25°C$
$2.00 atm = 50°C$ Temp doubles

18. $M = \frac{10.0}{102.89} = .09719$
$\approx 0.972 M$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Syntel Lopez **ID** 1011885
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

(1) $n = 1.00\,mol$
$V = 10.0\,L$
$T = 100.0^{\circ}C$
$P = ?$

(7) $P = 380\,mmHg \rightarrow atm$

(13) $m = \frac{mol\ solute}{kg\ solvent}$

$\frac{2.00\,mol}{0.500\,kg} = 4.00\,m$

(14) $M_1V_1 = M_2V_2$

$(1.00)_1(500mL) = (0.100)_2(500\,mL)_2$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Vanessa Arredondo **ID** 0618559
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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① $PV = nRT$ $R = 0.08206 \frac{L \cdot atm}{mol \cdot K}$

$\frac{10.0\,L \cdot 1}{1.00 \cdot 273.15}$ $\frac{10}{273}$

⑦ 380 mmHg atm = 760

$\frac{380}{1}$ 0.50

$C_1 = 1.00\,M$, $V_1 = 500.0\,mL$, $C_2 = 0.100\,M$

⑭ $m_1 V_1 = m_2 V_2$

$V_2 = \frac{(1.00)(500\,mL)}{0.100}$

Quantitative M/C SHOW-YOUR-WORK page. **Name**_____ **ID** _____
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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[5] Ideal gas $PV = nRT$

$n = 1.00 \text{ mol}$

$V = 10.0 \text{ L}$

$T = 100.0^\circ\text{C}$ $\quad 0.0821$

$$P = \frac{(1.00)(0.0821)(100.0)}{10.0}$$

[3]

380 mmHg to (atm) $\quad \frac{1}{760}$ $\quad$ (13) $\quad$ 20

$$380 \times \frac{1 \text{ atm}}{760 \text{ mmHg}}$$

$= 0.50 \text{ atm}$

[17]

$25.0 + 273.15 = 298 \text{ K}$

$P_1 = 1.00$

$T_1 = 25.0 \text{ K}$

$P_2 = 2.00$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\frac{1.00}{25.0} = \frac{2.00}{T_1}$$

[18]

$$M = \frac{\text{moles}}{\text{liters solution}}$$

$C_1 V_1 = C_2 V_2$

$(1.00)(V_1) = (0.100)(500.0)$ $\quad C_1 = 1.00$

$1.00 V_1 = 50$ $\quad V_2 = 500.0$

$\frac{1.00}{1.00} \quad \frac{50}{1.00}$ $\quad C_2 = 0.100$

50 $\quad V_1 = ?$

volumes mL-L
Temp °C - K
g-mol

1, 7, 13, 14, 17, 18

Quantitative M/C SHOW-YOUR-WORK page. **Name** Maria Gomez **ID** 0771844
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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Q1 $n = 1.00\,mol$ ideal gas
$V = 10.0\,L$ $\frac{PV}{V} = \frac{nRT}{V}$
$T = 100.0°C\ (100 + 273) = 373$
$P = ?$

$P = \frac{nRT}{V}$ $\frac{(1.00mo)(0.082058\ L\cdot atm)(373K)}{10.0L}$ $\frac{3.06607634}{10.0} = 3.0607\,atm$ **P = 3.06 atm**

Q7 Convert 380 mmHg to atm
1 atm → 760 torr
1 torr → 1 mmHg

$380\,mmHg \times \frac{1\,torr}{1\,mmHg} \times \frac{1\,atm}{760\,torr} = 0.5\,atm$

Q13 2.00 mol ($ZnCl_2$) dissolved in final 0.500 kg water.
molality of $ZnCl_2$?

$molality\ (m) = \frac{moles\ solute}{Kg\ solvent}$ $\frac{2.00\,mol}{0.500\,kg} = 4m$

Q14 dilution/concentration $(1.00m)(.500) = (0.100m)(500)$
dilution $C_1V_1 = C_2 \cdot V_2$ $V_1 = \frac{C_2 \cdot V_2}{C_1}$ $V_1 = \frac{0.100m(0.5L)}{1.00M} = 0.05L$
Known algebra? mL-L

$C_1 = 1.00M$
$V_2 = 500.0mL$
$C_2 = 0.100M$
$V_1 = unknown$

1000 mL = 1 L
$500.0mL \times \frac{1L}{1000mL} = 0.5L$

$0.05L \times \frac{1000mL}{1L} = 50.0mL$

Q17 Lussac $P \uparrow T \uparrow$ pressure double? Temp doubles?
$P_1 = 1.00\,atm$
$T_1 = 25.0°C + 273 = 298$
$P_2 = 2.00\,atm$
$T_2 =$

$P_2 \frac{P_1}{T_1} = \frac{P_2}{T_2} \cdot P_2$
$T_2 = \frac{P_1 \cdot P_2}{T_1}$ $T_2 = \frac{1.00atm(2.00atm)}{298}$ 0.0047

$\frac{P_1}{P_2 T_1} = \frac{P_2}{T_2} \cdot P_2 T_2$? 25+273
$T_2 = P_2 \times \frac{P_1}{T_1}$

Q18 $\frac{Sol\ (mol)}{Sol\ Vol\ (L)}$ g-mol mL-L
$10.0g \times \frac{1mol}{102.89g/mol} = 0.097$ $100.0mL \times \frac{1L}{1000mL} = 0.1L$

$\frac{0.097\,mol}{0.1L} = 0.97M$
0.972m

Quantitative M/C SHOW-YOUR-WORK page. **Name** Sabrina Garcia **ID** 04900 1
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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[3] $380\ mmHg \times \frac{1\ atm}{760\ mmHg} = 0.50\ atm$

[5] $\frac{(10.0)(100.0)}{0.082058\ \ 0.082058}$

[4] $2.00\ mol\ ZnCl_2 \times \frac{1}{136.29\ g/mol} = 1.467$

$0.500\ kg$

$\frac{2.00\ mol\ ZnCl_2}{0.500\ kg} = 4$

[3]

$\frac{(0.100\ M\ KCl)(500.0\ mL)}{1.00\ M\ KCL} = 50$

[4] 1p 1.00atm
Temp 25.0°C
↑ temp 2.00atm

$25.0 + 273.15 = 298.15$

[4] 10.0 g KBr 100.0 ML
Cmolar mass 102.89

$\frac{100.0\ mL}{102.89} = 0.9712$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Shane Xiong **ID** 1012185
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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4.) P of 380 mmHg to atm

$$380\ \text{mmHg} \times \frac{1\ \text{Torr}}{1\ \text{mmHg}} \times \frac{1\ \text{atm}}{760\ \text{Torr}} = \boxed{0.50\ \text{atm}}$$

8.) $P_1 = 1.00\ \text{atm}$ $T_1 = 25.0\ ^\circ\text{C}$ $P_2 = 2.00\ \text{atm}$ $T_2 = ?$ $\frac{P_1}{T_1} = \frac{P_2}{T_2} \rightarrow P_2 \cdot \frac{T_1}{P_1} = \frac{T_2}{P_2} \cdot P_2$

$$T_2 = \frac{2.00\ \text{atm} \times 25.0\ ^\circ\text{C}}{1.00\ \text{atm}} = \boxed{50\ ^\circ\text{C}} \qquad T_2 = \frac{P_2 T_1}{P_1}$$

9.) $C_1V_1 = C_2V_2$ $C_1 = 1.00\ \text{M KCl}$ $V_2 = 500.0\ \text{mL}$ $C_2 = 0.100\ \text{M KCl}$ $V_1 = ?$

$$V_1 = \frac{C_2V_2}{C_1} = \frac{0.100\ \text{M KCl} \times 500.0\ \text{mL}}{1.00\ \text{M KCl}} = \boxed{50.0\ \text{mL}}$$

13.) $PV = nRT$ $n = 1.00\ \text{mol}$ $V = 10.0\ \text{L}$ $T = 100.0\ ^\circ\text{C}$ $R = 0.082058\ \frac{\text{L atm}}{\text{mol K}}$ $P = ?$

$T = 100.0\ ^\circ\text{C} + 273.15 = 373.15\ \text{K}$

$$P = \frac{nRT}{V} = \frac{1.00\ \text{mol} \times 0.082058\ \frac{\text{L atm}}{\text{mol K}} \times 373.15\ \text{K}}{10.0\ \text{L}} = \boxed{3.06\ \text{atm}}$$

16.) Mass = 10.0g NaBr Molar Mass = 102.89 g/mol Volume = 100.0mL

$$M = \frac{n_{solute}}{V_{solution}} = \frac{0.0972\ \text{mol}}{100.0\ \text{mL}} \times 100 = \boxed{0.0972\ \text{M}}$$

$$\text{Moles} = \frac{\text{mass (g)}}{\text{molar mass } \frac{\text{g}}{\text{mol}}} = \frac{10.0\ \text{g NaBr}}{102.89\ \text{g/mol}} = 0.0972\ \text{mol}$$

19.) 2.00 mol $ZnCl_2$ Molar Mass = 136.29 g/mol 0.500 kg

$$m = \frac{n_{solute}}{\text{Kg solvent}} = \frac{2.00\ \text{mol}}{0.500\ \text{kg}} = \boxed{4.00\ \text{m}}$$

~~molar mass · Moles = Mass (g) / molar Mass~~ ~~· Molar Mass~~

$\text{Mass (g)} = \text{Molar mass } \frac{\text{g}}{\text{mol}} \cdot \text{Moles}$

$= 2.00\ \text{mol}\ ZnCl_2 \cdot 136.29\ \text{g/mol}$

$= 272.58\ \text{g}$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Samantha Gonzalez **ID** 0907899
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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1.) $P = \frac{nRT}{V}$ $P = \frac{1.00\,mol \cdot 0.082058\,L \cdot atm/(mol \cdot K) \cdot 373.15\,K}{10.0\,L} = 3.06199\,atm = 3.06\,atm$

$100.0^\circ C + 273.15 = 373.15\,K$

7.) $P = 380\,mmHg \cdot \frac{1\,atm}{760\,mmHg} = 0.5\,atm$

13.) $m = \frac{mol\ solute}{kg\ solvent} = \frac{2.00\,mol}{0.500\,kg} = 4.00\,m$

14.) $C_1 \cdot V_1 = C_2 \cdot V_2$ $\frac{C_1}{C_1}$ $V_1 = \frac{0.100\,M \cdot 500.0\,mL}{1.00\,M} = 50.0\,mL$

17.) $\frac{P_1}{T_1} = \frac{P_2}{T_2}$ $\frac{1.00\,atm}{25.0^\circ C} = \frac{2.00\,atm}{T_2}$ $T_2 = \frac{2.00\,atm \cdot 25.0^\circ C}{1.00\,atm} = 50.0^\circ C$

18.) $m = \frac{mol\ solute}{L\ solution}$ $10.0\,g \cdot \frac{1\,mol}{102.89\,g} = 0.09719\,mol$ $100.0\,mL \cdot \frac{1\,L}{1000\,mL} = 0.1000\,L$ $\frac{0.09719\,mol}{0.1000\,L} = 0.9719$

$M = 0.972$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Layla Moreno **ID** 10966633
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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4.) 380mmHg → atm

$380\ \text{mmHg} \times \frac{1\ \text{atm}}{760\ \text{mmHg}} = \boxed{0.50\ \text{atm}}$

8.) Gay-lussac

$P_2 = \frac{P_1(T_2)}{T_1} = 2.00\ \text{atm} = \frac{1.00\ \text{atm}(596.15\ \text{K})}{298.15\ \text{K}}$ ✓

$25 + 273.15 = 298.15\ \text{K}$

✓ $323 + 273.15 = 596.15\ \text{K}$

9.) $V_1 = \frac{C_2(V_2)}{C_1} = V_1 = \frac{0.100\ \text{M}(500.0\ \text{mL})}{1.00\ \text{M}} = \boxed{50\ \text{mL}}$

13.) $\frac{PV}{V} = \frac{nRT}{V} \rightarrow P = \frac{nRT}{V} = P = \frac{1.00\ \text{mol}(0.082058\ \text{L}\cdot\text{atm/mol}\cdot\text{K}(=373.15\ \text{K}))}{10.0\ \text{L}}$

$100^\circ\text{C} + 273.15 = 373.15\ \text{K}$

$= \boxed{3.06\ \text{atm}}$

16.) $M = \frac{\text{mol solute}}{\text{L solution}} = M = \frac{0.09719\ \text{mol}}{0.1\ \text{L}} = 0.97191 = \boxed{0.972}$

$\frac{10.0\ \text{g}}{102.89\ \text{g/mol}} = 0.09719\ \text{mol}$

$100.0\ \text{mL} \div 1000 = 0.1\ \text{L}$

19.) $m = \frac{\text{mol solute}}{\text{kg solvent}} = \frac{2.00\ \text{mol}}{0.500\ \text{kg}} = \boxed{m = 4}$

323.15

571.15 596.15

16 10 11 14 15 16 17 18 20

Quantitative M/C SHOW-YOUR-WORK page. **Name** Skye Lee **ID** 1073585

Indicate question with something like "Q9" for question 9, and demarcate your work on question.

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Q4 $380\ mmHg \times \frac{1\ atm}{760\ mmHg} = 0.50\ atm$ (STP)

Q8 $\frac{P_1}{T_1} = \frac{P_2}{T_2}$ $P_1 = 1.00\ atm$, $T_1 = 25.0°C$, $P_2 = 2.00\ atm$, $T_2 = ?$

$\frac{T_1 P_2}{P_1} = T_2$

$25.0°C + 273.15 = 298.15\ K$

$\frac{298.15\ K \times 2.00\ atm}{1.00\ atm} = ?\ K$

$= 596.3\ K - 273.15\ K = 323.15°C$ or $325°C$

Q9 $C_1V_1 = C_2V_2$

$C_1 = 1.00\ M\ KCl$, $V_1 = ?$, $C_2 = 0.100\ M\ KCl$, $V_2 = 500.0\ mL$

$V_1 = \frac{C_2V_2}{C_1}$

$\frac{0.100\ M\ KCl \times 500.0\ mL}{1.00\ M\ KCl} = 50.0\ mL$ (3 sig figs)

Q13 Ideal gas law

$PV = nRT$ $P = ?\ atm$, $V = 10.0\ L$, $n = 1.00\ mol$, $R = 0.08206\ \frac{L \times atm}{mol \times K}$, $T = 100.0°C$

$P = \frac{nRT}{V}$

$100.0°C + 273.15 = 373.15\ K$

$\frac{1.00\ mol \times 0.08206\ \frac{L \cdot atm}{mol \cdot K} \times 373.15\ K}{10.0\ L} = 3.0620689\ atm$

$= 3.06\ atm$ (3 sig figs)

Q19 $m = \frac{mol\ solute}{1\ kg\ solvent}$

$\frac{2.00\ mol\ ZnCl_2}{0.500\ kg\ water} = 4.00\ mol/kg$

$= 4.00\ m$

Q16

Quantitative M/C SHOW-YOUR-WORK page. **Name** Z Martinez **ID** 1004789
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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1) $PV = nRT$ $100°C + 273.15$ $T = 373.15\ K$

$P(10.0L) = (1.00\ mol)(0.082058\ \frac{L \cdot atm}{mol \cdot K})(373.15\ K)$ $R = (0.082058\ \frac{L \cdot atm}{mol \cdot K})$

$\frac{P(10.0L)}{10.0L} = \frac{(1.00)(0.082058\ atm)(373.15)}{10.0K}$ $V = 10.0L$ $n = 1.00\ mol$

$P = \frac{(1.00)(0.082058\ atm)(373.15)}{10.0} = 3.06\ atm$

7) $760\ mmHg = 1\ atm$

$380\ mmHg \times \frac{1\ atm}{760\ mmHg} = 0.50\ atm$

14) $V_1 = \frac{M_2 V_2}{M_1}$

$V_1 = \frac{(0.100)(500.0)}{1.00} = 50\ mL$

$25°C + 273.15 = 298.15K$

18) $M = \frac{mol}{L}$

$10.0_g \times \frac{1\ mol}{102.89_g} = 0.09719117$

$100\ mL = \frac{1\ L}{100\ mL} = 1\ L$

$\frac{0.0972}{1} = 0.0972\ M$

13) $m = \frac{mol}{kg\ solvent}$

$m = \frac{2.00}{0.500} = 4.00\ m$

17) $\frac{P_1}{T_1} = \frac{P_2}{T_2}$

$\frac{1.00}{298.15K} = \frac{2.00}{T_2}$

$0.003354016 = \frac{2.00}{T_2} \times 2.00$ $\times 2.00$

$0.00670833 = T_2$ -272.9929

$T_2 = -272.99°C + 25°C$

$T_2 = -297.9$

$-298°C$

17) ~~$\frac{P_1}{T_1} = \frac{P_2}{T_2}$~~ $25°C + 273.15$ 298.15

$T = \frac{(1.00\ atm)(298.15K)}{2.00\ atm} = 149.075K$

$149.075 - 273.15 = 124°C$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Marcella Perez **ID** 1011635
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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Q1 IGL $PV=nRT$

$P=$
$V=10.0\,L$
$n=1.00\,mol$
$T=100.0\,^{\circ}C = 373.15\,K$

$P(10.0\,L)=(1.00\,mol)(.082058\,\frac{L\,atm}{mol\,K})(373.15\,K)$
$P(10.0\,L)=30.61994127\,atm$
$P=3.061994127 = \lceil 3.06\,atm \rfloor$

Q7 mmHg -> atm
$1\,Torr = 1\,mmHg$
$1\,atm = 760\,Torr$

$380\,mmHg \times \frac{1\,Torr}{1\,mmHg} \times \frac{1\,atm}{760\,Torr} = .4871794871 = \lceil .5\,atm \rfloor$

Q14 $C_1V_1=C_2V_2$
$C_1=1.00\,M\,KCl$
$V_1=$
$C_2=.100\,M\,KCl$
$V_2=500.0\,mL$

$1.00\,M\,KCl(V_1)=(.100\,M\,KCl)(500.0\,mL)$
$1.00\,M\,KCl(V_1)=50.0\,M\,KCl\,mL$
$\lceil V_1=50.0\,mL \rfloor$

Q13 $molarity = \frac{mass\ as\ moles}{volume}$

$\frac{2.00\,mol\,ZnCl_2}{.500\,kg} = 4.00\,m\,ZnCl_2$

Q17 $\frac{P_1}{T_1}=\frac{P_2}{T_2}$
$P_1=1\,atm$
$T_1=25.0\,^{\circ}C$
$P_2=2.00\,atm$
$T_2=$

$\frac{1\,atm}{25.0\,^{\circ}C} = \frac{2.00\,atm}{T_2}$
$\frac{.04\,atm}{^{\circ}C} = \frac{2.00\,atm}{T_2}$

$\frac{.04\,atm}{^{\circ}C} \cdot T_2 = \frac{2.00\,atm}{1} \cdot \frac{^{\circ}C}{.04\,atm}$
$T_2 = 50\,^{\circ}C$

Q18

Quantitative M/C SHOW-YOUR-WORK page. **Name** Oluwapelumi Tay **ID** 1094793
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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(4) Conversion of 380 mmHg to atm

$\therefore 380\,\text{mmHg} \times \frac{1\,\text{atm}}{760\,\text{torr}} \times \frac{1\,\text{torr}}{1\,\text{mmHg}} = \frac{380\,\text{atm}}{760} = 0.50\,\text{atm}$

(8) $P_1 = 1.00\,\text{atm}$, $T_1 = 25.0^\circ\text{C}\,(298\,\text{K})$, $P_2 = 2.00\,\text{atm}$, $T_2 = ?$

$\frac{P_1}{T_1} = \frac{P_2}{T_2} \quad \therefore T_2 = \frac{P_2 T_1}{P_1} = \frac{2.00\,\text{atm} \times 298\,\text{K}}{1.00\,\text{atm}} = 596\,\text{K} = 323^\circ\text{C}$
(i.e $569\,\text{K} - 273\,\text{K}$)

(9) $C_1V_1 = C_2V_2$

$V_1 = \frac{C_2V_2}{C_1} = \frac{0.100\,\text{M} \times 500.0\,\text{mL}}{1.00\,\text{M}} = 50\,\text{mL}$

(13) $PV = nRT$

$n = 1.00\,\text{mol}$, $V = 10.0\,\text{L}$, $T = 100^\circ\text{C}\,(373\,\text{K})$ $R = 0.08205\,\text{atm}$

$P = \frac{nRT}{V} = \frac{1.00 \times 0.08205 \times 373}{10.0} = 3.06\,\text{atm}$

(16) Concentration of NaBr solution

$\text{mole} = \text{mass}/\text{molar mass} = 10.0\,\text{g}/102.89\,\text{g/mol} = 0.097\,\text{mol}$

$\text{Molarity} = \frac{\text{moles}}{\text{volume}} = \frac{0.097\,\text{mol}}{100\,\text{mL}} = 9.7 \times 10^{-4} \times 1000\,\text{L}$
(conversion of mL to L)
$= 0.972\,\text{M}$

(19) $\text{Molality} = \frac{\text{moles}}{\text{Kg}} = \frac{2.00\,\text{mol}}{0.500\,\text{Kg}}$

$= 4.00\,\text{m}$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Jennifer Ibarra **ID** 0873148
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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Q1) $P = n_{total} RT$

Q14) $C_1 = 1.00\ M$ was used to make $V_2 = 500.0\ mL$ of $C_2 = 0.100$

$$\frac{500}{10} = 50\ mL$$

Q17) $1.00\ atm = 25.0^\circ C$

$2.00 \times 25.0 = \boxed{50.0^\circ C}$

Q18) $\frac{100}{102.89} = 0.971\bar{9}$

$\boxed{0.972}$

Q7)

Quantitative M/C SHOW-YOUR-WORK page. **Name** Carlos Bedolla **ID** 1058710
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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7 $760/280 = .5$

14- $M_1 V_1 = M_2 V_2$

$1(500) = .1(?) = 50\ mL$

18- $n = \frac{mol}{L}$ $\frac{102.81}{10} = \frac{10,289\ mol}{.1\ L} =$

$\frac{190}{1000} = .1L$

29

Quantitative M/C SHOW-YOUR-WORK page. **Name** Lizett Castro **ID** 0991318
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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4.) 380 mmHg to atmosphress

380 mmHg

8) 1.00 Atm temp 25.0°C Increase → 2.00 Atm

$T_1 = 25.0 + 1.00 =$

$25.0 + 273.15 = 298.15$

25.0 1.00 Atm → 2.00 Atm

1.00 Atm

25.0 + 1.00 Atm

9) $C_1 = 1.00\ M\ KCl$ $V_2 = 500.0\ mL$

$C_2 = 0.100\ MCL$

19) 136.29 ÷ 0.500kg = 272.58

272.58 ÷ 2.00 = 139.29

16) 10.0g 102.89 100mL

102.89 ÷ 100.0 mL = 1.0289

Quantitative M/C SHOW-YOUR-WORK page. **Name**_____ **ID** _____
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

1°

Quantitative M/C SHOW-YOUR-WORK page. **Name** Jose Sanchez **ID** 10854163
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

4. $380 \text{ mmHg} \times \frac{1 \text{ atm}}{1 \text{ mmHg}} = 380 \text{ atm}$

8. Everything constant as Pressure inc temp dec $\frac{25^\circ C}{2} = 12.5^\circ C$

9 $V_2 = 500 \text{ mL}$ $\frac{(1.00 \text{ atm}) \times (?)}{.100 \text{ atm}}$

$\frac{(1.00 \text{ atm})(50.0 \text{ mL})}{.100 \text{ atm}} = 500 \text{ mL}$

13 $P = \frac{(1.00 \text{ mol})(0.0805 \frac{\text{L atm}}{\text{mol K}})(273 + 100^\circ C)}{10. \text{ L}}$

$= 3.06 \text{ atm}$

16 $10.0 \text{ g} \times \frac{1 \text{ mol}}{102.89} \times \frac{1}{100 \text{ mL}} \times \frac{1000}{1 \text{ L}} = 0.972 \text{ M}$

19 $m = \frac{\text{mole Solute}}{\text{kg Solvent}} = \frac{2.00 \text{ mol}}{.500 \text{ kg water}}$

$= 4.00 \text{ m}$

Quantitative M/C SHOW-YOUR-WORK page. **Name** JUSTIN VINK **ID** 1070339
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

(4)

$380\ mmHg = \frac{1\ atm}{760\ mmHg} = 0.5$

$760\ mmHg = 1\ atm$

(16) $10.0 / 102.89 = 0.097191175$

0.1

$100\ mL = 0.1\ L$

0.097191175

(8) $\frac{P_1}{T_1} = \frac{P_2}{T_2}$ $25 + 273.15$

~~$\frac{1\ atm}{298.15} \times \frac{2\ atm}{F_2}$~~

$\frac{1\ atm}{25^\circ} = \frac{2\ atm}{T_2}$

50

~~$596.3 - 273.15$~~

(19) $2.00 / 136.29 =$

[illegible]

0.500

[illegible]

2.00

0.500

$= 4.00$

(9) $\frac{(1.00)(V_1)}{1.00} = (0.100)(500.0)$

$\frac{50}{1.00} = 50.0$

(13) $\frac{PV}{V} = \frac{nRT}{V}$ $R = 0.08206$

$P = (1.00)(0.08206)(373.15) = \frac{30.620689}{10.01}$

3.0620689

Quantitative M/C SHOW-YOUR-WORK page. **Name** Emily M Galicia **ID** 1028935
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

Quantitative M/C SHOW-YOUR-WORK page. **Name** Natalie Ochoa **ID** 09518655
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

None

Quantitative M/C SHOW-YOUR-WORK page. **Name** Alyssa Martinez **ID** 0931469
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

Q 9: $V_1 = \frac{C_2 V_2}{C_1}$

$$V_1 = \frac{(0.100\ M\ KCl)(500.0\ mL)}{1.00\ M\ KCl} \qquad \frac{50\ mL}{1.00\ M\ KCl} = 50\ mL \checkmark$$

Q 13:

$n = 1.00\ mol \quad V = 10.0\ L \quad T = 100.0^{\circ}C$

?

Q 16: $10.0\ g\ NaBr \times \frac{1\ mol\ NaBr}{102.89\ g\ NaBr} \times \frac{1}{100.0\ mL} \times \frac{1000\ mL\ solution}{1\ L\ solution} =$

0.9719

$\approx 0.972\ M \checkmark$

Q 19: ? 2.00 m

Quantitative M/C SHOW-YOUR-WORK page. **Name** Isaac Mack **ID** 0936354
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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$PV = nRT$

(1)

$$\frac{P(10.0)}{10.0} = \frac{(1.00)(0.08206)(373.15)}{10.0}$$

$P = 3.0620689$

(7) (or 280 torr)

$$\frac{380 \text{ mmHg}}{1} \times \frac{1 \text{ atm}}{760 \text{ mmHg}}$$

$= 0.5 \text{ atm}$

(13) $m = \frac{\text{mols of solute}}{\text{kg of solvent}}$

$\frac{2.00 \text{ mol}}{0.500 \text{ kg}} = 4.00 \text{ m}$

(14) $C_1V_1 = C_2V_2$

$$\frac{(1.00)V_1}{1} = \frac{(0.100)(500.0)}{1}$$

$V_1 = 50.0 \text{ mL}$

(17) $P_1/T_1 = P_2/T_2$

$(1.00)/(298.15) = (2.00)/T_2 \times \frac{1}{2.00}$

$\times \frac{1}{2.00}$

$(1.00)/(298.15) \times \frac{1}{2.00} = T_2$

$1/25 = 2/T_2$

$\times T_2 \quad \times T_2$

$1/25 = 2 \times 25$

$\times 25$

$T_2 = 50$

this took an embarrassing amount of time 💀

(18) $M = \text{mol}/L$

$$\frac{10.0 \text{ g}}{1} \times \frac{1 \text{ mol}}{102.89 \text{ g}} = 0.097191175$$

$$\frac{100.0 \text{ mL}}{1} = \frac{0.001 \text{ L}}{1 \text{ mL}} = 0.1 \text{ L}$$

$$\frac{0.097191175 \text{ mol}}{0.1 \text{ L}} = 0.9716 \text{ M}$$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Cynthia Sanchez **ID** 1093672
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

13. $n = 1.00\ mol$ $V = 10.0\ L$ $T = 100.0 + 273.15\ ^\circ C$

$$\frac{10.0 \cdot atm}{373.15} = 3.06$$

$$8.25 \times 2 = 50.\ ^\circ C$$

$373.15 = 1.00 \times$

$$25.0 + 273.15 = 298.15$$

$$\frac{L \cdot atm}{mol \cdot k}$$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Robert D Ruiz **ID** 1075374
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

Q4.) 380 mmHg to atm

1 atm = 760 torr
1 torr = 1 mm Hg

$?\ atm = \frac{1\ atm}{760\ torr} \cdot \frac{380\ mmHg}{1} = 0.5$

Q8.) 1.00 atm at 25.0°C (298.15 K) to 2.00 atm ~~$\frac{PV}{nt}$~~ Guy-Lussac

0°C = 273.15 K

?°C =

Q9.) 1.00 M KCl makes 500.0 mL (.5 L) 0.100 M solution

$C_1 \cdot V_1 = C_2 \cdot V_2$ (in mol/L & L)

$1.00\ M\ KCl \cdot ?\ L\ KCl = .5\ L \cdot 0.100\ M$

$?\ L = \frac{.5\ L \cdot .1\ M}{1\ M} = .05\ L = 50\ mL$ (× 1000 mL)

Q13.) Calculate ideal gas Pressure (n = 1.00, V = 10.0 L, T = 100.0°C) $PV = nRT$

$?P = \frac{1\ mol \cdot (.082058\ L \cdot atm/mol \cdot K) \cdot 373.15\ K}{10.0\ L}$ T = 373.15 K $P = \frac{nRT}{V}$

?P = 3.06199427 atm

Q16.) Concentration of NaBr molar mass = 102.89 g/mol, 10.0 g NaBr, 100.0 mL solution

molarity = mol solute / 1 liter solution

$?\ mol\ NaBr = 10.0\ g\ NaBr \cdot \frac{1\ mol\ NaBr}{102.89\ g} = .097191175\ mol$

$?\ M = .097191175\ mol\ NaBr / 0.1\ L = 0.972\ M$

Q19.) molality of solution 2.00 mol $ZnCl_2$, .5 kg water 136.29 g/mol molality = $\frac{mol\ solute}{Kg\ solvent}$

~~? mol $ZnCl_2$ =~~ $\frac{2.00\ mol}{0.50\ kg} = 1m$ 1.00 m

Quantitative M/C SHOW-YOUR-WORK page. **Name** Aaron Robles **ID** 0955832
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

1. $PV = nRT$

$(P)(10L) = (1)(R)(100.0C)$

7. $380 mmHg \times \frac{1 atm}{760 mmHg} \quad \frac{1}{2} = 0.500 atm$

17. $P_1 = 1 atm \quad P_2 = 2 atm$
$t_1 = 25°C \quad t_2 =$

$\frac{1 atm}{25°C} = \frac{2 atm}{t_2}$

$\frac{50 atm \cdot C}{1 atm} = \frac{1 atm (t_2)}{1 atm}$

$50°C = t_2$

18. $\frac{10 g\ NaBr}{102.89} = 0.097 mol\ NaBr$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Guisela Gallardo **ID** 0577380
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

4) $1\ atm = 760\ mmHg$ $380\ mmHg \times \frac{1\ atm}{760\ mmHg} = 0.5\ atm$

8) $\frac{P_1}{T_1} = \frac{P_2}{T_2}$ $\frac{1.00\ atm}{25.0°C} = \frac{2.00\ atm}{T_2}$ $= T_2 = \frac{2.00\ atm \times 25.0°C}{1.00\ atm} = 50°C$

9) $C_1V_1 = C_2V_2$ $\frac{(1.00\ M\ KCl)V_1}{1.00\ M\ KCl} = \frac{(0.100\ M\ KCl)(500.0\ mL)}{1.00\ M\ KCl} = 50\ mL$

13) $PV = nRT$ $P(10.0\ L) = (1.00\ mol)\left(0.08205 \frac{L \cdot atm}{mol\ K}\right)(373°K)$ $\frac{273 + 100°C}{373°K}$

$P = \frac{30.60465}{10.0\ L} = 3.060465\ atm$

16) $M = \frac{102.89\ g\ NaBr \times \frac{1\ mol\ NaBr}{}}{0.1\ L}$

19) $m = \frac{mol\ solvent}{kg\ solvent}$ $\frac{2\ mol}{0.500\ kg} = 4.00\ m$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Carina Mendoza **ID** 0718748
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

Q15 $\frac{(500.0)(0.100)}{1.00} = 50.0\ mL$

Q1 $25 + 273 = 298\ K$

$\frac{(1.00)(0.0821)(298)}{100.0}$

$n = 1.00\ mol$

volume $V = 10.0\ L$

temp $T = 100.0\ C$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Fernanda Dominguez **ID** 0981154
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

1) $PV = nRT$

P? $V = 10.0\,L$

$T = 100.0°C + 273.15 = 373.15K$

$n = 1.00\,mol$

$P(10.0L) = (1.00)(0.082058)(373.15)$

$\frac{P(10.0)}{10.0} = \frac{30.6199427}{10.0}$

$= 3.0619\ atm$

7 $P = \frac{F}{A}$

$380\,mmHg \left(\frac{1\,atm}{760\,mmHg}\right)$

$= \frac{1}{2}$ or $0.50\ atm$

13)

$m = \frac{moles\ solute}{kg\ solvent}$

$m = \frac{2.00}{0.500} = 4.00\,m$

14) $C_1V_1 = C_2V_2$

$C_1 = 1.00\,mKC$

$V_1 = ?$

$C_2 = 0.100\ MKCl$

$V_2 = 500.0\,mL$

$\frac{(1.00)}{1}V = (0.100)(500.0)$

$V = \frac{50}{1} = V = 50\,mL$

17) $\frac{P_1}{T_1} = \frac{P_2}{T_2}$

$P_1 = 1.00\,atm$

$T_1 = 25.0°C + 273.15 = 298.15$

$P_2 = 2.00$

T_2?

$\frac{1.00}{298.15} = \frac{2.00}{T_2}$

$2.00 \times 0.003354016 = \frac{2.00 \times 2.00}{T_2}$

$= 323°C$

18)

$10.0g\ NaBr$

$100.0\,mL$

$m = \frac{10.00\,(102.89)}{100.0} =$

$=$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Ariana Sutton **ID** 1011137
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

1. $PV = nRT$ $n = 1.00\ mol$, $V = 10.0\ L$ $T = 100.0°C$

$P(10.0L) = (1.00mol)R(100.0°C)$

3.06 atm

7. 380 mmHg to atm

$380\ mmHg \times \frac{1\ atm}{760} = \frac{380\ mmHg}{760} = 0.5\ atm$

13. $molality(m) = \frac{mole\ solute}{kg\ solvent}$

$molality(m) = \frac{2.00\ mol}{0.500\ kg} = 4.00\ m$

14. $C_1 \times V_1 = C_2 \times V_2$

$1.00\ M\ KCl \times V_1 = 0.100\ M\ KCl \times 500.0\ mL$

$\frac{1.00\ M\ KCl \times V_1}{1.00} = \frac{50.0\ mL}{1.00} = 50.0\ mL$

17. $\frac{1.00\ atm}{25.0°C} = \frac{2.00\ atm}{T_2}$

I forgot what equation to use

18. $M = \frac{102.89\ g/mol}{100.0}$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Mandy Yang **ID** 0869542
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

(1) Given: $n = 1.00\ mol$
$V = 10.0\ L$
$T = 100.0^\circ C$
$R = 0.08205\ \frac{L \cdot atm}{mol \cdot K}$
convert: $^\circ C \rightarrow K$
$100.0^\circ C + 273K = 373\ K$
Finding: P
setup: $P = \frac{nRT}{V}$
calculation:
$P = \frac{(1.00\ mol)(0.08205\ \frac{L \cdot atm}{mol \cdot K})(373K)}{(10.0L)}$
$= 3.06\ atm$

(7) converting mmHg → atm
Given: 380 mmHg
Finding: atm
1 atm = 760 mmHg
$380\ mmHg \times \frac{1\ atm}{760\ mmHg} = 0.50 atm$

(13) Given: 2.00 mol of $ZnCl_2$
mm of $ZnCl_2 = 136.29\ g/mol$
0.500 kg
Finding: m
setup: $m = \frac{mol\ solute}{kg\ solvent}$
calculation:
$m = \frac{2.00\ mol\ ZnCl_2}{0.500\ kg} = 4.0\ m$

(14) Given: $C_1 = 1.00\ M\ KCl$
$V_2 = 500.0\ mL$
$C_2 = 0.100\ M\ KCl$
Finding: $V_1 = ?\ mL$
setup: $V_1 = \frac{C_2 V_2}{C_1}$
calculations:
$V_1 = \frac{0.100\ M \times 500.0\ mL}{1.00\ M} = 50.0\ mL$

(17) Given: $P_1 = 1.00 atm$
$T_1 = 25^\circ C$
$P_2 = 2.00\ atm$
Finding: $T_2 = ?\ K$
setup: $T_2 = \frac{P_1 T_1}{P_2}$
calculations:
$T_2 = \frac{1.00\ atm \times 25^\circ C}{2.00\ atm} = 12.5^\circ C$

(18) Given: 10.0 g NaBr
mm = 102.89 g/mol
100.0 mL
Finding: M
setup: $M = \frac{mol}{L}$
converting: mL → L
100.0 mL ÷ 1000 = 1100 L
102.89 g/m × 10.0 = 1028.9
calculations:
$M = \frac{1028.9}{1100L} = 0.972\ M$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Gabriella Ochoa **ID** 1035210
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

ex- 1 atm = 760 mmHg
760 atm = 760 mmHg

4) pressure (p) of 380 mHg to atm
mmHg → atm 380 mmHg = 380 atm

8) pressure (p) - 1.00 atm → 2.00 atm
temp (T) - 25.0°C → 50.0°C

50.00°C → k
50.00 → + 273.15 = 323 K

9) given: $C_1 = 1.00$ M KCl (moles)
$= V_2 = 500.0$ mL of a $C_2 = 0.100$ M KC
$V_1 = ?$

19) 2.00 mol zinc
solvent - does dissolving
solute - gets dissolved

molality = $m = \frac{\text{mol solute}}{\text{kg solvent}}$

$= \frac{2.00 \text{ mol zinc}}{0.500 \text{ kg}}$

13) $n = 1.00$ mol
$V = 10.0$ L → =
$T = 100.0°C \rightarrow 373.15$
+ 273.15
p ? atm

$pV = nRT$

$p = \frac{1.00(0.08206)(373.15)}{V = 10.0 \text{ L}}$

$= 3.06$ atm

16) 10.0 g NaBr (102.89 g/mol)
100.0 mL in water
10 x 102.89 g/mol = 1028.9 g/mol

$0°C = 273°K$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Charlie Hatch **ID** 0992103
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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Q13.) $n = 1.00\ mol \quad V = 10.0\ L \quad T = 100.0°C \quad R = 0.082058\ L \cdot atm/(mol \cdot K)$

$\frac{3 \cdot 10.0}{1.00 \cdot 100}$ $nRT = PV$

$1.00 \cdot 0.082058 \cdot 373°K = 30.6\ atm$

Q4.) $380\ mm\ Hg \times \frac{1\ atm}{760\ mm\ Hg} = 0.50\ atm$

Q8.) $1.00\ atm \rightarrow 2.00\ atm$
$T = 25.0°C \rightarrow 50.00°C$
according to Gay-Lussac's law temp should increase linearly with pressure

Q9.) $C_1 V_2 = V_1$
$0.100 \cdot 500.0 = 50.0\ mL$

Q16.) $10.0\ g\ NaBr \cdot \frac{1\ mol\ NaBr}{102.89\ g/mol\ NaBr} = 0.0972\ mol\ NaBr$

$100.0\ mL \cdot \frac{1\ L}{1000\ mL} = 0.10\ L \quad \frac{.0972}{.10} = 0.972\ M$

Q19.) $\frac{2.00\ mol\ Zn}{0.500\ kg\ H_2O} = 4.00\ molality$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Allen Benitez **ID** 1049064
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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(4)

$$380\ \text{mmHg} \left| \frac{1\ \text{atm}}{760\ \text{mmHg}} \right. = \boxed{0.50\ \text{atm}}$$

(8)

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\frac{1.00\ \text{atm}}{25^\circ\text{C}} = \frac{2.00}{T_2}$$

$$(1.00\ \text{atm})(25^\circ\text{C}) = (2.00\ \text{atm})(T_2)$$

$$T_2 = \boxed{12.5^\circ\text{C}}$$

(8)

$$T_2 \left(\frac{1.00\ \text{atm}}{25^\circ\text{C}} \right) = 2.00\ \text{atm}$$

$$\frac{(1.00\ \text{atm})\,T_2}{25^\circ\text{C}} = (2.00)(25^\circ\text{C})$$

$$\frac{(1.00)\,T_2}{1.00\ \text{atm}} = \frac{(2.00\ \text{atm})(25^\circ\text{C})}{1.00\ \text{atm}} = \boxed{50.0^\circ\text{C}}$$

(19)

$$\frac{2.00\ \text{mol}}{0.500\ \text{kg}} = 4.00\ \frac{\text{mol}}{\text{kg}}$$

(9)

$$C_1V_1 = C_2V_2$$

$$(1.00\ \text{M KCl})(V_1) = (0.100\ \text{M KCl})(500.0\ \text{mL})$$

$$V_1 = \boxed{50.0\ \text{mL}}$$

(13)

$$\frac{PV}{V} = \frac{nRT}{V}$$

T: C to K
$100^\circ\text{C} + 273.15 = 373.15\ \text{K}$

$$P = \frac{(1.00\ \text{mol})\left(0.082058\ \frac{\text{L·atm}}{\text{mol·K}}\right)(373.15\ \text{K})}{(10.0\ \text{L})}$$

$$P = \boxed{3.06\ \text{atm}}$$

(16)

$$M = \frac{\text{mol}}{\text{L}} = \frac{0.0972\ \text{mol}}{0.1\ \text{L}} = \boxed{0.972\ \text{M}}$$

$$10.0\ \text{g NaBr} \left| \frac{1\ \text{mol}}{102.89\ \text{g/mol}} \right. = 0.0972\ \text{mol NaBr}$$

$$100.0\ \text{mL} \left| \frac{1\ \text{L}}{1000\ \text{mL}} \right. = 0.1\ \text{L}$$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Donna Adam **ID** 1111240
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

1. $V = 10.0\,L$ $PV = nRT$
$n = 1.00\,mol$
$T = 100.0^\circ C$ $P\,10.0\,L = 1.00\,mol \cdot 100.0^\circ C$

14. $V_2\ 500.0\,mL = \frac{1.00\,M\,KCl}{0.100\,M\,KC} = 5.00\,mL$

17. $1.00\,atm \times \frac{25.0^\circ C}{2.00\,atm} = 12.5^\circ C$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Nadia Diaz **ID** 09896716
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

$1.00\,atm \times 25.0^{\circ}C \div 2.00\,atm = 12.5^{\circ}C$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Mina Schulte **ID** 0934994
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

~~7) $P = P_1V_1 = P_2V_2$~~

1) $PV = nRT$ $r = 0.8206$

$P = \frac{nRT}{V}$ $n = 1.00$ $V = 10.00L$ $T = 100^\circ C$

$100 + 273.15 = 373.15$

$P = \frac{(1.00)(0.8206)(373.15)}{10.00L} = 30.620689$

$= 30.6$ atm

13) 2.00 mol

136.29 g/mol - 0.500kg

$\frac{2.00\,mol}{0.500kg} = 4.00m$

17.) $P = 1.00$ atm

$t = 25.^\circ C$

$P_1 t_1 = P_2 t_2$

$t_2 = \frac{1.00\,(25.0)}{2.00} = 12.5^\circ C$

18) 10g 102.89 g/mol 100.0ml

$\frac{10g}{102.89\,g/mol} = 0.097191175$

7) 380 mmHg → atm

1 mol gas = 22.4L at stp

stp = 0°C, 1.0 atm $\frac{380}{22.4} = 16.96$

?

14) $m_1V_1 = m_2V_2$

~~$V = \frac{(1.00)(500.0)}{0.100}$~~ 500

$\frac{(0.100)(500.0)}{1.00} = \frac{50}{} = 50.0ml$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Cecilia Kangiser **ID** 1084367
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

4) 380 mmHg → atm 0.50

1 atm = 760 Torr

8) 1.00 atm = 25.0°C

2.00 atm = 50.9°C

9) mL ex. 00.0 ← one decimal place

13)

16)

19)

Quantitative M/C SHOW-YOUR-WORK page. **Name** Rosa M Contreras **ID** 09152236
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
THIS and ALL sheets must be turned in with your choice responses, whether used or unused

1) $n = 1.00\ mL$ $V = 10.0\ L$ Temp $100.0^\circ C$

$P = \frac{nRT}{V}$ $\frac{(1)(0.0821)(373)}{10}$

6) $Mg(s) + Cu(NO_3)_2(aq) \rightarrow Mg(NO_3)_2(aq) + Cu(s)$

7) $760 = Torr$

8) H Halogen

10) $BaCl_2 + Na_2SO_4 \rightarrow$

12) water solvent
Salut Solvet

13) 2.00 mol
0.500 kg

~~136.29~~ $m = \frac{m}{L}$ $m = \frac{2.00\ mol}{1\ kg}$ Kg -> L

1000g ->

500g

14) $m_1 V_1 = m_2 V_2$

$\frac{500.0 \cdot 0.100}{1}$

17) $P_1 = \frac{P_2 T_1}{T_2}$

$\frac{(2)(25)}{1}$ $\frac{(2)(1)}{25}$

18) 10.8g V_2 100.0 mL

m_1 102.89

10.0g 1100.0g

17) P_1 1.00 atm T_1 °C 25.0 P_2 2.00

K
273 + 25 = 298

$\frac{(1)(298)}{2}$

$T_2 = \frac{P_1 \cdot T_1}{P_2}$

$T_2 =$ 149

V
$102.89 \cdot 10 = m_2\ 1000$

0.9353

Quantitative M/C SHOW-YOUR-WORK page. **Name** Abel Figueroa **ID** 0831812
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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(4) $380\,mmHg \quad \frac{1\,atm}{760\,mmHg} = \boxed{0.5\,atm}$

(8) $\frac{1.00\,atm}{(273+25\,^{\circ}C)K} = \frac{2.00\,atm}{T_2} \qquad T_2 = \frac{2.00\,atm\,(298.15\,K)}{1.00\,atm} = 596.3\,K - 273\,K = \boxed{323.15\,^{\circ}C}$

(9) $\frac{100}{x}\ V_1\ KCl \qquad \frac{500.0\,L\ KCl \cdot 100\,M\,KCl}{1.00\,M\,KCl} = \frac{500.0\,L\ KCl \cdot 100\,M\,KCl}{0.100\,M\,KCl} = 5000\,L \times \frac{.01\,mL}{1\,L} = 50.0\,mL$

(13) $P = \frac{(1.00\,mol)(0.082056\,\frac{L\cdot atm}{mol\cdot K})(273+100\,^{\circ}C)K}{(10.0\,L)} \qquad 373.15$

$= \frac{30.6199427}{10.0\,L} = \boxed{3.06\,atm}$

(16) $10.0\,g\ NaBr = 100.0\,mL$ of NaBr in water Concentration?

Molar Mass
$102.89\,g/mol$

$\frac{10\,g}{102.89\,g/mol} = 0.0972\,mol$ Solute

$\boxed{0.0972\,mol}$

$100.0\,mL \times \frac{1\,L}{0.01\,mL} = \frac{10{,}000\,L}{22.4\,L} = 446.42\,mol$

(19) $2.00\,mol\ ZnCl_2 \qquad 0.500\,kg$ water

Molar mass
$136.29\,g/mol$

$M = \frac{2.00\,mol\ ZnCl}{0.500\,kg} = \boxed{4.00\,M}$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Emily G Mavsesian **ID** 0822239
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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1. Ideal gas
$Pv = nRT$ $\quad n = 1.00\ mol$ $\quad R = 0.082058$
$\frac{nRT}{n}$ $\quad V = 10.0L$
$T = 10.0$

$$\frac{(1.00\ mol)(0.082058)(10.00)}{100.0\ Mol} = 0.820\ sd \quad 0.2248392$$

$$Pv = nRT$$

~~P.~~ 7.) $P(380\ mmHg) = 380\ mmHg \times \frac{1\ atm}{760\ torr} = 0.5$

17. Gas = P = 1.00 atm, Temp = 25.0 °C Pressure
↑ to 2.00 atm.
$P = 1atm$ $\quad Temp = 25.0°C$ $\quad V = kTn$ $\quad 25 \times \frac{1\ atm}{2.00} =$

18. 10 g of sodium (NaBr) Molar Mass = 102.89 g/mol
is used to Make 100.0 ML in
what. whats the NaBr

$$10\ g \cdot \frac{10.00}{100.0\ ML} = \frac{1}{0.9} \rightarrow 1$$

14. $C_1 = 1.00$
$V_2 = 500.0$
$C_2 = 0.100\ M =$ $\quad \frac{(1.00)(500.0}{0.100} = 50$

7.

P

Quantitative M/C SHOW-YOUR-WORK page. **Name** Ezaballa Garcia **ID** 0913233
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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4.) $380\text{mmHg} \times \frac{1\text{ atm}}{760\text{mmHg}} = 0.50\text{ atm}$

8.) $2.00 - 1.00 = 1.00\text{ atm}$

9.) $C_1 = 1.00\text{ M HCl}$ $C_2 = 0.100\text{ M HCl}$ $V_2 = 500.0\text{mL}$

$V_2 = \frac{1.00 \times 0.100}{500.0\text{ mL}} = 0.0002 \times 0.001$

13.) $\frac{PV}{V} = \frac{nRT}{V}$

$100.0^\circ\text{C} + 273.15 =$

$P = \frac{1.00\text{mol}\left(0.0820578\text{ L·atm/mol·K}\right) \times 373.15\text{ K}}{10.0\text{ L}} = 3.06\text{ atm}$

16.) $10.0 \times \frac{100.0}{102.89} = 9.7191\ldots/100 = 0.0972$

19.)

Quantitative M/C SHOW-YOUR-WORK page. **Name** Martin Garcia **ID** 0681538
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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$PV = nRT$

$100.0 + 273.15 = 373.15\,K$

1. $P = \frac{nRT}{V}$ $\frac{(1.00\,mol)(0.082058\,L\,atm/mol\,K)(373.15\,K)}{10.0\,L} = 3.06144427\,atm$

7. $380\,mm\,Hg = 380\,Torr$

$\frac{380\,Torr}{760\,Torr/atm} = .5\,atm$

13. $\frac{2.00\,mol}{.5\,kg} = 4.00\,m$

14. $\frac{C_1V_1}{C_1} = \frac{C_2V_2}{C_1}$ $\frac{0.100\,M(500.0\,mL)}{1.00\,M} = 50\,mL$

$V_1 = \frac{C_2V_2}{C_1}$

17. $\frac{P_1T_1}{P_2} = \frac{P_2T_2}{P_2}$ $\frac{(1.00\,atm)(298.15\,K)}{2.00\,atm}$

$T_2 = \frac{P_1T_1}{P_2}$

18. $\frac{10.0\,g}{102.89\,g/mol} = 0.09719175\,M$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Alexander Ruiz **ID** 10913670
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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1. $n = 1.00\ mol$ $V = 10.0\ L$ $Temp = 100.0^\circ C$ Find P

$PV = nRT \Rightarrow P = \frac{nRT}{V} \Rightarrow P = \frac{(1.00)(0.306)(100.0^\circ C)}{(10.0\ L)} = 3.06\ atm$

7. $P = 380\ mmHg \rightarrow ATM$

$380\ mmHg \times \frac{1\ Torr}{1\ mmHg} \times \frac{1\ atm}{1\ Torr} = 380\ atm$

13. 2 mol Zn Molar mass = 136.29 g/mol dissolve in final 0.500 kg

$M = \frac{2.00}{0.500} = 4.00\ m$

14. $C_1 = 1.00\ M\ KCl$ $V_2 = 500.0\ mL$

$C_2 = 0.100\ M\ KCl$

$C_1V_1 = C_2V_2 \Rightarrow V_2 = \frac{C_2V_2}{C_1} = \frac{(0.100)(500)}{1.00} = 50.0\ mL$

16. NO_2

$x + 2(-2) = 0$

$x + (-4) = 0$

$x = +4$

17. P_1 T_1 P_2 T_2
IP: 1.00 atm + Temp = 25.0 → 2.00 atm Temp ?

$\frac{P_1}{T_1} = \frac{P_2}{T_2} = \frac{(25.0)(2.0)}{(1.00)} = \frac{50.0}{1.00} = 50.0^\circ C$

18. 10.0 g of NaBr g/mol = 102.89 -> 100.0 mL

V_2

$C_1V_1 = C_2V_2$

$V = 100.0\ mL \times mass = 102.89 = 10289$

$V = \frac{mol}{m}$

Quantitative M/C SHOW-YOUR-WORK page. **Name** Galilea Leon **ID** 0904104
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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(13) $n = 1.00\ mol$ $\quad PV = nRT \rightarrow P = \frac{nRT}{V}$
$V = 10.0\ L$
$T = 100.0°C$ $\quad P = \frac{(1.00)(0.082058)(373.0)}{10.0}$
$P = ?$
$100 + 273 = 373.0\ C°$ $\quad = 3.06037634$
≈ 3.06

(8) $P_1 = 1.00\ atm$ $\quad P_2 = 2.00$
$T_1 = 25.0C$ $\quad T_2 = ?$
$\frac{P_1}{T_1} = \frac{P_2}{T_2}$ $\quad T_2 = \frac{P_1 T_1}{P_2}$ $\quad T_2 = \frac{(1.00)(25.0)}{2.00} = 12.5\ °C$

(16) $Mg + Cu(NO_3)_2 \rightarrow Mg(NO_3)_2 + Cu$
swapped

(19) $2.00\ mol\ (ZnCl_2)$
[molar mass = 136.29 g/mol]
$0.500\ kg$ water (solvent)
$\text{molality} = \frac{\text{moles solute}}{\text{kg solvent}}$
$m = \frac{2.00}{0.500} = 4\ m$

(4) $P = 380\ mmHg \rightarrow atm$
$1\ atm = 760\ mmHg = 760\ torr$
$380\ mmHg \times \frac{1\ atm}{760\ mmHg} = \frac{1}{2} = 0.50\ atm$

(9) $C_1 = 1.00\ M\ KCl$
$V_2 = 500.0\ mL$
$C_2 = 0.100\ M\ KCl$
$V_1 = ?$
$C_1V_1 = C_2V_2$
$V_1 = \frac{C_2V_2}{C_1}$
$V_1 = \frac{(0.100)(500.0)}{1.00} = 50\ mL$

16. 10.0g NaBr 100.0mL

[molar mass = 102.89 g/mol]

$M = \frac{n\text{Solute}}{V\text{ solution}}$ $M = \frac{102.89}{100.0\text{mL}}$ or $\frac{10.0}{100.0\text{mL}}$?

$\frac{g}{mol}$ $\frac{10.0g}{102.89}$ $= 0.09719\ldots \div 100.0\text{mL}$

≈ 0.0972?

Quantitative M/C SHOW-YOUR-WORK page. **Name** Nhia Lee **ID** 0884241
Indicate question with something like "Q9" for question 9, and demarcate your work on question.
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3. $380 \text{ mmHg} \times \frac{1 \text{ atm}}{760 \text{ mmHg}}$

$\frac{380}{760} = 0.5 \text{ atm}$

8. $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

$T_2 = V_2 \frac{T_1}{V_1}$ $2.00 \times \frac{(25.0c + 273)}{1.00 \text{ atm}}$

$2.00 \times \frac{298}{1.00} = 596 - 273$

$323^\circ C$

9. $C_1 V_1 = C_2 V_2$

$V_1 = \frac{C_2 V_2}{C_1}$ $\frac{0.100 \times 500.0}{1.00} = 50 \text{ mL}$

13. $PV = nRT$

$P = \frac{nRT}{V}$ $P = \frac{1.00 (0.0206)(100 + 273)}{10.0 L}$

$\frac{273}{100}$

$P = \frac{1.00 (0.0206)(373)}{10.0}$

$= 3.060$

3.06

16. $\frac{10.0}{102.89} = 0.09719$

19. molality = $\frac{\text{mass}}{\text{kg}}$

$\frac{2.00}{136.29} \quad \frac{136.29}{0.500}$ $\qquad$ $\frac{2.00}{0.500\text{ kg}} \quad \frac{1000\text{ g}}{1\text{ kg}}$

$\frac{2.00}{0.5000} = 4.00$

Quantitative M/C SHOW-YOUR-WORK page. **Name**_____ **ID** _____
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$P_1 = 1.00$ atm $\quad P_2 = 2.00$ atm

$T_1 = 25.0$ $\quad T_2 = ?$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$T_2 = \frac{P_2 \times T_1}{P_1} = \frac{2.00 \times 25}{1.00}$$

$$= 50.0$$

$$C_1 V_1 = C_2 V_2$$

$C_1 = 1.00$ $\quad C_2 = 0.100$

$V_1 = ?$ $\quad V_2 = 500.0$

$$V_1 = \frac{C_2 V_2}{C_1}$$

$$V_1 = \frac{0.100 \times 500.0}{1.00}$$

$$V_1 = 50$$

$$x + 2(-2) = 0$$

$$x - 4 \quad x = 4$$

$$4 - 2.4$$

$$m = \frac{2.00}{0.500}$$

$PV = nRT$

$P = \frac{nRT}{V}$

$T = 100.C$

$K = 100 + 273.15$

373.15

$P = \frac{1.00 \times 0.0820 \times 373.15}{10.0L}$

$\frac{10.0}{102.89}$

$\frac{100 mL}{1000}$

0.1